

Evaluation of Labor Market Openness Variable Construction

Global Capitalism Index Subindex Review

I. Executive Summary

The Labor Market Openness subindex represents an ambitious attempt to operationalize labor market characteristics within a capitalist framework. While the theoretical grounding is generally sound and draws appropriately from classical and contemporary economic thought, significant concerns exist regarding data quality, conceptual validity of certain measures, and the operationalization of labor migration. The reliance on World Economic Forum (WEF) survey data for the majority of indicators raises questions about measurement consistency and comparability across countries and time periods.

II. Evaluation of Theoretical Framework

Strengths

The theoretical framework provides a coherent narrative linking labor market openness to capitalist functioning. The authors effectively integrate classical perspectives (Locke, Marx, Polanyi) with modern labor economics (Becker) and macroeconomic theory (Keynesian and neoliberal models). The tripartite conceptual structure—Labor Market Flexibility, Labor Protections, and Labor Migration—represents a reasonable decomposition of the broader concept.

The recognition that human capital requires free deployment analogous to physical or financial capital is well-founded. The inclusion of anti-discrimination and equality considerations as fundamental to market efficiency, rather than merely normative concerns, demonstrates sophisticated theoretical thinking.

Concerns

1. Conceptual Tension: There is an unresolved tension between "flexibility" (which often implies minimal regulation) and "protections" (which require regulation). The framework does not adequately explain how these potentially competing values are balanced or weighted in the overall construct.

2. Missing Elements: The framework omits several important dimensions of labor market openness:

- Occupational licensing and professional credential recognition
- Non-compete agreements and other contractual restrictions on mobility
- Monopsony power and labor market concentration
- Informal sector dynamics, which are critical in many developing economies
- Digital labor platforms and gig economy considerations

3. Schumpeterian Interpretation: The application of creative destruction to labor markets may be theoretically questionable. While firm turnover drives innovation, rapid labor turnover can undermine firm-specific human capital investment and reduce productivity.

Conclusion: The theoretical framework is adequate but incomplete. The model makes sense as far as it goes, but several vital elements of labor market openness in contemporary capitalism are missing.

III. Component Validity Analysis

A. Labor Market Flexibility Component

Datasets Included:

- WEF Hiring and Firing Practices
- WEF Cooperation in Labor-Employer Relations
- WEF Ease of Hiring Foreign Labor
- WEF Flexibility of Wage Determination
- WEF Pay and Productivity

Assessment:

This component relies entirely on WEF Executive Opinion Survey data. While these measures capture important perceptions of labor market flexibility, they suffer from several limitations:

1. Survey Consistency Problems: The WEF survey methodology has changed over time, with variations in sample composition, response rates, and question wording. The reliability scores (3.575-4.075) reflect moderate confidence at best.

2. Elite Bias: The WEF survey predominantly samples business executives and economic elites. Their perceptions of "flexibility" may systematically differ from workers' experiences and may overemphasize employer interests.

3. Conceptual Overlap: Several measures appear to capture similar underlying constructs (e.g., hiring/firing flexibility and ease of hiring foreign labor both relate to employment regulation).

Missing Indicators:

- OECD Employment Protection Legislation indices (more objective measures of regulatory stringency)
- World Bank Doing Business indicators for hiring practices (discontinued but available historically)
- Actual labor turnover rates and job-to-job transition data
- Measures of temporary/fixed-term contract prevalence
- Active labor market policy expenditures

B. Labor Protections Component

Datasets Included:

- ILO Compliance with Labor Rights
- V-Dem Freedom From Forced Labor
- WEF Wage Inequality Between Men and Women
- WEF Inclusion of Women in Leadership

Assessment:

This is the strongest component in terms of data quality. The ILO and V-Dem sources are highly regarded for their rigor and cross-national comparability.

Strengths:

- ILO data (Grade 4.75) represents systematic monitoring of international labor standards with transparent methodology
- V-Dem data (Grade 5.0) benefits from expert surveys with careful aggregation and uncertainty estimates
- The inclusion of both freedom from forced labor and gender equality represents important dimensions of labor market openness

Weaknesses:

- Only two WEF indicators here, but they still introduce perceptual bias
- Missing comprehensive measures of:
 - Child labor prevalence
 - Workplace safety regulations and enforcement
 - Minimum wage adequacy
 - Union density and collective bargaining coverage
 - Workplace discrimination beyond gender (race, ethnicity, disability, age)

Data Quality: This component has the most defensible data sources. The ILO is indeed a terrific source, and V-Dem's rigorous methodology provides confidence in their measures.

C. Labor Migration Component

Datasets Included:

- World Bank Net Migration
- WEF Internal Labor Force Mobility
- WEF Fill Vacancies by Hiring Foreign Labor
- WEF Ease of Finding Skilled Labor in Local Labor Market

Assessment:

Critical Conceptual Problems:

This component fundamentally fails to measure what it purports to measure. The indicators are, at best, weak proxies for labor migration openness:

1. Net Migration Is Not Migration Policy: Net migration reflects the outcome of many factors (economic conditions, conflict, climate, demographic trends) and tells us almost nothing about a country's openness to labor migration. A country could have restrictive immigration policies but positive net migration due to refugee flows, or open policies but negative net migration due to poor economic conditions.

2. Perception Measures Are Not Policy Measures: Three of four indicators rely on WEF survey perceptions. "Fill vacancies by hiring foreign labor" and "internal labor force mobility" measure perceived labor market tightness and domestic mobility patterns, not migration policy openness.

3. Local Labor Market Skills: The fourth WEF indicator (ease of finding skilled labor) measures human capital endowment and education systems, not migration policy.

Missing Critical Indicators:

- OECD Migration Policy Indices (MIPEX) - comprehensive measures of integration policies
- Visa policies and work permit regulations
- Recognition of foreign qualifications and credentials
- Language and integration requirements
- Pathway to permanent residence and citizenship for workers
- Temporary labor migration quotas and programs

- Bilateral labor agreements
- Remittance flows (as an outcome measure)
- Refugee and asylum policies as they relate to labor market access

Recommendation: This component requires complete reconceptualization. If the goal is to measure labor migration openness, direct policy measures are essential. If such measures are unavailable, this component should be renamed to reflect what it actually measures (e.g., "Labor Market Outcomes" or "Perceived Labor Availability").

IV. Data Source Quality Assessment

A. World Economic Forum Executive Opinion Survey

Concerns:

- 1. Methodological Inconsistency:** The WEF survey has undergone multiple revisions in sampling strategy, weighting procedures, and question formulation over the 2008-2023 period. This creates time series comparability problems.
- 2. Sample Representativeness:** The survey systematically oversamples business elites in capital cities and major economic centers. Rural areas, smaller economies, and diverse economic sectors may be underrepresented.
- 3. Response Bias:** As a survey of executives, responses may reflect employer preferences rather than objective labor market conditions. Questions about "flexibility" may be interpreted through the lens of employer interests.
- 4. Perceptual vs. Objective Measures:** Survey data captures perceptions, which may lag or lead actual policy changes and may be influenced by recent economic shocks or media narratives.
- 5. Redundancy with Existing WEF Indices:** As noted, the heavy reliance on WEF data essentially recreates components of the WEF Global Competitiveness Index. This raises questions about the added value of this exercise.

Reliability Scores: The reliability grades assigned to WEF indicators (3.575-4.075) reflect moderate confidence, yet these indicators dominate the overall index (9 of 14 indicators are from WEF).

B. International Labour Organization (ILO) Data

Strengths:

The ILO Compliance with Labor Rights indicator represents high-quality data from a specialized international organization with strong technical capacity and country presence. The ILO's methodology is transparent, consistently applied, and based on treaty monitoring and reporting obligations. This is an excellent data source that should be expanded if additional ILO indicators are available.

Potential Additional ILO Data:

- Statistics on informal employment
- Working poverty rates
- Collective bargaining coverage
- Occupational injury and fatality rates

C. V-Dem (Varieties of Democracy) Data

Strengths:

V-Dem represents best-practice expert survey methodology with multiple safeguards against bias. The forced labor indicator is well-suited to this index, and V-Dem's transparent uncertainty estimates add analytical value.

Potential Additional V-Dem Data:

- Worker autonomy indicators
- Freedom of association measures (beyond ILO)
- State capacity for labor law enforcement

D. World Bank Data**Assessment:**

Net migration data from the World Bank is of reasonable quality (Grade 5.0) but, as discussed above, does not measure migration policy openness. The World Bank also produces Doing Business indicators (now discontinued) that would have been more appropriate for labor market regulation.

V. Methodological Evaluation**Principal Components Analysis Approach****Strengths:**

- PCA provides data-driven weighting rather than arbitrary assignment
- The layered approach (components → subindices → composite index) allows modularity
- Reliability scoring integrated into the weighting process is sophisticated

Concerns:

1. Assumption of Unidimensionality: PCA assumes that variation in the component indicators reflects variation in a single underlying dimension (labor market openness). However, the theoretical framework suggests multiple dimensions that may not align along a single axis.

2. Year-Specific Normalization: Normalizing data within each year prevents meaningful comparison of absolute levels across time. This means the index measures relative position within a year rather than change in actual openness over time. A country could maintain constant policies but see its score change due to shifts in other countries.

3. Missing Data Handling: While the clustering approach to handle missing data is reasonable, it may create discontinuities in time series when country coverage changes.

4. First Principal Component Dominance: The approach assumes the first principal component adequately captures the construct. However, if multiple dimensions exist, substantial information may reside in higher-order components.

Data Harmonization

The min-max normalization approach using year-specific ranges creates comparability problems:

- A score of 5.0 in 2010 may represent different absolute conditions than 5.0 in 2020
- This makes it difficult to assess whether countries are converging or diverging in actual policies

- Temporal trends may reflect changes in sample composition rather than real policy shifts

VI. Recommendations for Improvement

Immediate Priorities

1. Reduce WEF Dependence: The current index is essentially a weighted average of WEF survey perceptions with some additional data. Either:

- Replace WEF indicators with more objective policy measures where possible
- Clearly label this as a "Perceived Labor Market Openness" index
- Commission independent expert surveys using V-Dem methodology for labor market characteristics

2. Reconceptualize Labor Migration Component: Completely redesign this component to include direct measures of migration policy or eliminate it if such data are unavailable. Current measures do not validly capture the theoretical construct.

3. Expand ILO and V-Dem Data Usage: These are the strongest data sources. Conduct a comprehensive review of available indicators from these sources and expand their use.

4. Address Informal Sector: Most labor in developing economies operates informally. The absence of informal sector measures creates serious validity concerns for cross-national comparison.

Additional Data Sources to Consider

- **OECD Employment Protection Legislation Indices:** Objective measures of regulatory stringency
- **OECD Migration Policy Indices (MIPEX):** Comprehensive migration policy data
- **ILO ILOSTAT Database:** Additional labor market indicators (informal employment, working poverty, etc.)
- **UN Population Division:** Migration stock and flow data
- **International Trade Union Confederation (ITUC) Global Rights Index:** Union rights and collective bargaining
- **US State Department Human Rights Reports:** Forced labor and trafficking assessments
- **Walk Free Foundation Global Slavery Index:** Modern slavery prevalence estimates

Conceptual Additions

- **Labor Market Concentration:** Measures of employer market power and monopsony
- **Occupational Licensing:** Barriers to occupational mobility
- **Non-Compete Agreements:** Contractual restrictions on labor mobility
- **Gig Economy Regulation:** Platform work and worker classification
- **Informal Sector Size:** Share of employment outside formal regulation
- **Labor Force Participation:** Gender and age disaggregated rates
- **Skills Mismatch:** Over-qualification and under-employment

VII. Required Additional Analyses and Reporting

To properly evaluate the robustness and validity of the Labor Market Openness subindex, the following analyses and reporting are required:

A. Measurement Uncertainty and Standard Errors

Requirement: Report standard errors or confidence intervals for all measures at multiple levels:

- 1. Component-Level Standard Errors:** For each of the three components (Labor Market Flexibility, Labor Protections, Labor Migration), provide bootstrapped standard errors or confidence intervals for country-year scores.
- 2. Subindex-Level Standard Errors:** Report uncertainty estimates for the overall Labor Market Openness subindex score for each country-year observation.
- 3. Data Source Uncertainty:** Where available (particularly for V-Dem data), incorporate source-provided uncertainty estimates into the PCA process and propagate these through to final scores.
- 4. Temporal Stability:** Calculate year-over-year changes in scores with confidence intervals to distinguish meaningful change from measurement noise.

Justification: Without understanding measurement uncertainty, it is impossible to assess whether observed differences between countries or changes over time are statistically meaningful or simply reflect measurement error and sampling variation.

B. Correlation Matrix of Component Indicators

Requirement: Provide a comprehensive correlation matrix showing:

- 1. Within-Component Correlations:** Correlation matrices for all indicators within each component:
 - All five WEF indicators in Labor Market Flexibility
 - All four indicators in Labor Protections
 - All four indicators in Labor Migration
- 2. Cross-Component Correlations:** Correlations between indicators across different components to identify potential redundancy or conceptual overlap.
- 3. Temporal Stability of Correlations:** How correlation patterns change over time (e.g., do indicators become more or less aligned over the 2008-2023 period?).

Justification: High within-component correlations may indicate redundancy (multiple measures of the same concept), while low correlations may suggest that indicators are measuring different constructs and should not be combined. Cross-component correlations help validate the theoretical separation of the three components.

C. Component-Subindex Correlation Analysis

Requirement: Report correlations between:

- 1. Individual Indicators and Component Scores:** For each indicator, report its correlation with its component's overall score.
- 2. Component Scores and Subindex Score:** Correlation between each of the three component scores (Labor Market Flexibility, Labor Protections, Labor Migration) and the overall Labor Market Openness subindex score.
- 3. Component Weight Distribution:** Document the empirically derived PCA weights assigned to each component in generating the subindex score.

Justification: These correlations reveal whether components are contributing independently to the overall construct or whether one component dominates the subindex. They also help

identify which specific indicators are driving component scores versus those that contribute marginal information.

D. Country Coverage Analysis

Requirement: Provide detailed reporting on:

1. Country Coverage by Year: For each year from 2008-2023, report:

- Total number of countries with complete data
- Number of countries with partial data
- Number of countries with missing data
- List of countries entering or exiting the sample

2. Indicator-Specific Coverage: For each of the 14 indicators, report:

- Number of countries covered per year
- Geographic distribution of coverage
- Income-level distribution of coverage (low, lower-middle, upper-middle, high income)

3. Missing Data Patterns: Document systematic patterns in missingness:

- Are certain regions consistently underrepresented?
- Do smaller or poorer countries have less complete data?
- How does missing data cluster (which indicators tend to be missing together)?

4. Coverage Implications: Analyze how changes in sample composition over time may affect score comparability and interpretation.

Justification: Systematic patterns of missing data can introduce selection bias. If coverage varies by region, income level, or regime type, the index may not be comparable across these groups. Changes in coverage over time can create spurious trends.

E. Temporal Variation Analysis

Requirement: Document how Labor Market Openness scores vary over time:

1. Aggregate Trends: Overall distribution of subindex scores across all countries for each year (mean, median, standard deviation, range).

2. Country-Specific Trajectories: Identify:

- Countries with largest positive changes in scores over the period
- Countries with largest negative changes in scores over the period
- Countries with most stable scores
- Countries with high year-to-year volatility

3. Component-Level Trends: Track how each of the three components changes over time:

- Is Labor Market Flexibility increasing or decreasing globally?
- Are Labor Protections strengthening or weakening?
- How do perceived Labor Migration patterns evolve?

4. Policy Event Analysis: For selected countries with major policy changes (e.g., labor law reforms, migration policy shifts), document whether and how quickly scores respond to these known changes.

5. Correlation Between Year-Specific Scores: Analyze temporal autocorrelation to assess whether scores are stable or volatile year-over-year.

Justification: Understanding temporal variation is essential for:

- Distinguishing real policy changes from measurement artifacts
- Assessing whether the min-max normalization obscures absolute changes
- Identifying whether certain components are more volatile than others
- Validating that the index responds appropriately to known policy reforms

VIII. Conclusion

The Labor Market Openness subindex represents a valuable attempt to systematize measurement of an important dimension of capitalist economies. The theoretical framework is generally sound, though it could be strengthened by addressing conceptual tensions and incorporating additional dimensions of contemporary labor markets.

The primary weakness lies in data source selection. Over-reliance on WEF Executive Opinion Survey data introduces systematic measurement concerns related to elite bias, perceptual rather than objective measurement, and temporal inconsistency. The Labor Migration component, in particular, requires fundamental reconceptualization, as current indicators are weak proxies at best for migration policy openness.

The strength of the index lies in its incorporation of ILO and V-Dem data, which represent high-quality, rigorously collected information. These sources should be expanded wherever possible, and additional objective policy measures should replace or supplement WEF survey data.

The methodological approach using PCA is sophisticated but introduces interpretive challenges, particularly regarding temporal comparability due to year-specific normalization. The required additional analyses outlined in Section VII are essential to assess the index's validity, reliability, and appropriate interpretation.

With significant revision to the Labor Migration component, reduction in WEF dependence, and comprehensive reporting of uncertainty and correlations, this subindex could provide valuable insights into labor market openness as a dimension of capitalist systems. In its current form, it should be interpreted cautiously, particularly for temporal comparisons and for the migration dimension.